

Math 110 – Homework 2 Answers

Math 110 – Fall 2026

Homework 2

Linear independence, span, bases, and dimension

Suggested completion date: Friday, September 11

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. (LADR §2A, Exercise 11.) Prove or give a counterexample: if $v_1, \dots, v_n$ and $w_1, \dots, w_n$ are linearly independent lists in V , then

$$v_1 + w_1, \dots, v_n + w_n$$

is linearly independent.

Problem 2. Let $\mathbb{F}^\infty$ be the vector space of all sequences

$$(a_1, a_2, a_3, \dots), \quad a_j \in \mathbb{F},$$

with coordinatewise addition and scalar multiplication. Let e_j denote the sequence with 1 in the j th coordinate and 0 elsewhere.

- (a) Verify that $\mathbb{F}^\infty$ is a vector space over $\mathbb{F}$.
- (b) Prove that $\mathbb{F}^\infty$ is infinite-dimensional.
- (c) Prove that $e_1, e_2, e_3, \dots$ do *not* span $\mathbb{F}^\infty$.

Problem 3. (LADR §2B, Exercise 5.) Suppose V is finite-dimensional and U, W are subspaces such that $V = U + W$. Prove that V has a basis all of whose vectors belong to $U \cup W$.

Problem 4. Let $\mathbb{F}$ be a field.

- (a) Prove that every nonzero proper subspace of $\mathbb{F}^2$ has the form $\text{span}(v)$ for some nonzero $v \in \mathbb{F}^2$.
- (b) Let $v, w \in \mathbb{F}^2$. Prove that v, w form a basis of $\mathbb{F}^2$ if and only if neither vector is a scalar multiple of the other.

Problem 5. (LADR §2C, Exercise 12.) Suppose U, W are subspaces of $\mathbb{R}^8$ such that

$$\dim U = 3, \quad \dim W = 5, \quad U + W = \mathbb{R}^8.$$

Prove that $\mathbb{R}^8 = U \oplus W$.

Problem 6. Define

$$U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}.$$

- (a) Prove that U is a subspace of $\mathcal{P}_4(\mathbb{R})$.
- (b) Find a basis of U and determine $\dim U$ directly, without using rank-nullity.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes
